

1. Cancer cells survive therapy and regrow after treatment during relapse by:
 - a. Attracting microorganisms
 - b. Secreting sticky substances
 - c. Attacking the immune system
 - d. Expressing multidrug-resistant proteins, thus providing resistance to many chemotherapeutic agents
2. The ability of adult stem cells to form specialized cell types of tissues other than the place where they reside is known as:
 - a. Plasticity
 - b. Translation
 - c. Transplantation
 - d. Transduction
3. Red and white blood cells are derived from which type of the following stem cells?
 - a. Kidney stem cells
 - b. Stromal cells
 - c. Hematopoietic cells
 - d. Liver stem cells
4. In double trisomy, the haploid number of chromosomes is repeated three times.
 - a. True
 - b. False
5. The vector that can be used to transfer 80-90 kb of DNA is:
 - a. Yeast chromosome
 - b. Multicopy plasmid
 - c. Bacteriophage P1
 - d. Bacteriophage lambda
6. Which of these is mismatched?
 - a. Infertility — Inability to conceive after 1 year of unprotected and regular sexual intercourse
 - b. Primary infertility — Couples have never had children
 - c. Secondary infertility — Couples initiated conception in the past and then had difficulty

- d. Primary infertility — Couples initiated conception in the past and then had difficulty
- 7. In humans, aneuploidy includes all the following EXCEPT:
 - a. Diploidy
 - b. Monosomy
 - c. Trisomy
 - d. Triploidy
- 8. Point mutation can be detected by which of the following techniques?
 - a. IUI
 - b. Polymerase chain reaction (PCR)
 - c. ICSI
 - d. FISSI
- 9. In human females, progesterone (P4) is necessary for pregnancy maintenance.
 - a. True
 - b. False
- 10. To generate cultures of specific types of differentiated cells—heart muscle cells, blood cells, for example—scientists try to control differentiation of embryonic stem cells by all the following methods EXCEPT:
 - a. Change the chemical composition of the culture medium
 - b. Changing the temperature of the cell culture medium
 - c. Modify the cells by inserting specific genes or adding specific growth factors
 - d. Alter the surface of the culture dish
- 11. Cybrids result from fusion between:
 - a. Cells depleted in mitochondrial DNA (mtDNA) and synaptosomes from aged mice, which often contain mutated mtDNA
 - b. Egg and differentiated cells
 - c. Cancer stem cells and normal cells
 - d. Egg and sperm
- 12. Embryonic stem cells from a donor introduced into a patient could cause transplant rejection.
 - a. True
 - b. False

13. Epithelial adult stem cells in the lining of the digestive tract occur in deep crypts and give rise to the following cell types EXCEPT:

- a. Follicle cells
- b. Absorptive cells
- c. Entero-endocrine cells
- d. Goblet cells

14. The procedure for introducing exogenous donor DNA into a recipient cell is called:

- a. Transfection
- b. Transportation
- c. Transplantation
- d. Translation

15. IVF procedure includes all the following EXCEPT:

- a. Removal of the female pronucleus from the embryo
- b. Insemination
- c. Superovulation
- d. Luteal support

16. Embryonic stem cells (ESC) line is established when all the following are reached EXCEPT:

- a. ESC appear genetically normal
- b. ESC that have become pluripotent
- c. Proliferating ESC in tissue culture for two weeks without differentiating
- d. Proliferating ESC in cell culture for six or more months without differentiating

17. Triploid embryos can be diagnosed by which of the following techniques?

- a. PCR
- b. FISH
- c. Electrophoresis
- d. IUI

18. Which of these is mismatched?

- a. Pluripotent cells — Inner cell mass
- b. Totipotent cell — Zygote

- c. Totipotent cells — Inner cell mass
- d. Totipotent cells — Cells of the four-cell stage embryo

19. Which step of culturing human embryonic stem cells (ESC) contains a wrong statement?

- a. The inner surface of the culture dish is typically coated with a feeder layer
- b. The reason for having the mouse feeder cells in the bottom of the culture dish is to give the inner cell mass cells protection from microbes
- c. Over the course of several days, the ESC proliferate and begin to crowd the culture dish
- d. Human embryonic stem cells are isolated by transferring the inner cell mass into a plastic laboratory culture dish that contains a nutrient broth known as culture medium

20. Which of these is NOT considered a psychological harm of reproductive cloning in humans?

- a. To avoid homozygosity in case of recessive alleles (parents are carriers)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
D	A	C	B	B	B	D	A	D	B	A	B	D	B	A	B	D	A	C	B